Given the report indicator and row number, each simulated data set is replicable. If you have questions, please contact Zhengxiao Wei (dextersherlo@gmail.com). Feb. 13, 2023

btw_dat.csv

80,000 obs. of 8 variables for between-subjects designs (Figures 2 and 3)

- **BF** - Bayes factor value, computed by the *anovaBF* function (v0.9.12; Morey & Rouder, 2022)

- **SEP.CI** - separation percentage of the standard confidence intervals (in %)
  Note: contain negative values. So do the other types of SEP%.
  `SEP% = 50 * (mean.Level1 - mean.Level2) / CI.width`

- **SEP.HPDI** - separation percentage of the standard highest (posterior) density intervals (in %)

- **BFerror** - Bayes factor estimate error, computed by the *anovaBF* function (v0.9.12; Morey & Rouder, 2022)
  Note: in a range of 3.545437e-22 to 6.890953e-04.

- **Report** - report indicator for archived Table 2
  — eight levels: Report1, Report2, Report3, Report4, Null24, Null48, Null100, Report5

  Note: Report1 to 4 are for archived Table 2. Null24/48/100 are for simulations under the null with particular sample sizes. Report5 is the extreme cases simulated under Cohen's *d* = 2.5.

- **Effect.Size** - true effect size used for simulations
  — four levels: Small, Large, Null, Extreme

- **Sample.Size** - number of subjects in a balanced group
  — three values: 24, 48, 100

- **Case** - case indicator for Table 2
  — three levels: Case1, Case2, None

- **BF_Pearson** - Pearson Bayes factor value
  Note: shape parameter is set to 0 in Faulkenberry (2021).

wi_dat.csv

180,000 obs. of 14 variables for within-subject designs (Figures 2, 4, 5, and 6)

- **BF** - Bayes factor value, computed by the *anovaBF* function (v0.9.12; Morey & Rouder, 2022)

- **SEP.LM.CI** - separation percentage of the confidence intervals by Loftus and Masson (1994) (in %)

- **SEP.NKM.HDI** - separation percentage of the highest-density intervals by Nathoo, Kilshaw, and Masson (2018) (in %)

- **SEP.LH.HDI** - separation percentage of the highest-density intervals by Ly et al. (2017) and Heck (2019) (in %)

- **SEP.anova.HDI** - separation percentage of the highest-density intervals computed by the *anovaBF* function in Appendix B (in %)

- **SEP.JZS.HDI** - separation percentage of the highest-density intervals computed by the *rmHDI* function by Wei, Nathoo, and Masson (v.0.1.15; 2022) (in %)

- **SEP.HPDI** - separation percentage of the highest (posterior) density intervals (in %)

- **BFerror** - Bayes factor estimate error, computed by the *anovaBF* function (v0.9.12; Morey & Rouder, 2022)

  Note: in a range of 0.004759049 to 0.006126002.

- **Report** - report indicator for archived Table 3

  — eighteen levels: Report1, Report2, Report3, Report4, Report5, Report6, Report7, Report8, Report9, Report10, Report11, Report12, Null1, Null2, Null3, Null10, Null11, Null12

  Note: Report1 to 12 are for archived Table 3. Null 1 to 3 and 10 to 12 are for simulations under the null with report indicators.

- **Effect.Size** - true effect size used for simulations

  — three levels: Large, Small, Null

- **Rho** - correlation between the two conditions

  — three values: 0.9, 0.7, 0.3

- **Effect.Size…Rho** - combined factor of Effect.Size and Rho

- **Sample.Size** - number of subjects

  — two values: 24, 48

- **Case** - case indicator for Table 3

  — five levels: Case1, None, Case2, Case3, Case4

vary_dat.csv

330,000 obs. of 8 variables for within-subject designs (Figure 7)

- **BF** - Bayes factor value, computed by the *anovaBF* function (v0.9.12; Morey & Rouder, 2022)

- **SEP.JZS.HDI** - separation percentage of the highest-density intervals computed by the *rmHDI* function by Wei, Nathoo, and Masson (v.0.1.15; 2022) (in %)

  Note: contain NA values because some simulations were not conducted. In some simulations, SEP% has taken the absolute value, may resulting in mismatched sign with SEP.anova.HDI.

- **SEP.anova.HDI** - separation percentage of the highest-density intervals computed by the *anovaBF* function in Appendix B (in %)

  Note: contain NA values because some simulations were not conducted. In some simulations, SEP% has taken the absolute value, may resulting in mismatched sign with SEP.JZS.HDI.

- **BFerror** - Bayes factor estimate error, computed by the *anovaBF* function (v0.9.12; Morey & Rouder, 2022)

  Note: in a range of 0.004359703 to 0.892087533.

- **Effect.Size** - true effect size used for simulations

  — five values: 0, 3.8, 6.7, 12.0, 20.4

- **Sample.Size** - number of subjects

  — five values: 5, 10, 24, 36, 48

- **Rho** - correlation between the two conditions

  — two values: 0.9, 0.3

- **Case** - case indicator for Table 3

  — two levels: Case1, Case4

vary_dat.btw.csv

150,000 obs. of 7 variables for between-subjects designs (Figure 7 supplementary)

- **BF** - Bayes factor value, computed by the *anovaBF* function (v0.9.12; Morey & Rouder, 2022)

- **SEP.CI** - separation percentage of the standard confidence intervals (in %)

- **SEP.HPDI** - separation percentage of the standard highest (posterior) density intervals (in %)

- **BFerror** - Bayes factor estimate error, computed by the *anovaBF* function (v0.9.12; Morey & Rouder, 2022)
  Note: in a range of 1.857768e-15 to 4.604857e-04.

- **Effect.Size** - true effect size used for simulations
  — three levels: Small, Large, Null

- **Sample.Size** - number of subjects
  — five values: 24, 48, 5, 10, 36

- **Case** - case indicator for extended Table 2
  — five levels: Case1, Case2, Case3, Case4, Case5

Extended Table 2

Parameters for simulated data in a between-subjects design with two conditions

Case	1	2	3	4	5
n	24	48	5	10	36
μ_1	100	100	100	100	100
μ_2	(100, 91.5, 83.5)	(100, 94.1, 88.4)	(100, 79.4, 59.5)	(100, 86.4, 73.5)	(100, 93.1, 86.6)
σ_ϵ	20	20	20	20	20
power	(NA, .3, .8)	(NA, .3, .8)	(NA, .3, .8)	(NA, .3, .8)	(NA, .3, .8)

Table 3

Parameters for data simulation in a within-subject design with two conditions

Case	1	2	3	4
n	24	24	48	48
μ_b	100	100	100	100
$\Delta\mu$	(0, 3.8, 6.7)	(0, 16.5, 28.6)	(0, 2.7, 4.6)	(0, 12.0, 20.4)
σ_b	20	20	20	20
σ_ϵ	6.67	30.55	6.67	30.55
ρ	.9	.3	.9	.3

Archived Table 2

Parameters for simulated data in a between-subjects design with two conditions

Report	1	2	3	4
n	24	48	24	48
μ_1	100	100	100	100
μ_2	91.5	94.1	83.5	88.4
σ_ϵ	20	20	20	20
power	.3	.3	.8	.8

Archived Table 3

Parameters for data simulation in a within-subject design with two conditions

Report	1	2	3	4	5	6	7	8	9	10	11	12
n	24	24	24	24	24	24	48	48	48	48	48	48
μ_b	100	100	100	100	100	100	100	100	100	100	100	100
$\Delta\mu$	6.7	13.1	28.6	3.8	7.5	16.5	4.6	9.0	20.4	2.7	5.4	12.0
σ_b	20	20	20	20	20	20	20	20	20	20	20	20
σ_ϵ	6.67	13.10	30.55	6.67	13.10	30.55	6.67	13.10	30.55	6.67	13.10	30.55
ρ	.9	.7	.3	.9	.7	.3	.9	.7	.3	.9	.7	.3
power	.8	.8	.8	.3	.3	.3	.8	.8	.8	.3	.3	.3